

MOGOCHA, Russian Federation

WMO: 306730

Lat: 53.750N

Lon: 119.733E

Elev: 2050

StdP: 13.64

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.9	-37.3	-44.5	0.4	-40.3	-40.9	0.6	-36.9	18.6	-5.6	15.7	-3.9	0.2	140	0.671

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.5	83.1	65.0	79.8	63.6	76.6	62.3	68.0	77.3	66.1	75.3	64.3	72.9	5.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	99.1	72.0	63.0	92.8	70.0	61.1	86.9	68.2	33.7	76.6	32.1	75.3	30.6	72.9	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.2	14.2	11.8	DB	-45.5	88.5	4.5	2.9	-48.8	90.6	-51.4	92.3	-53.9	93.9	-57.2	96.0	(4)
(5)			WB	-45.6	71.3	4.4	2.2	-48.7	72.9	-51.3	74.2	-53.7	75.4	-56.9	77.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.0	-16.6	-7.7	9.5	30.4	45.8	58.7	63.5	58.5	44.7	26.3	0.2	-15.9	
	DBStd	30.05	8.78	9.67	11.63	8.44	6.91	6.08	4.82	5.89	7.36	9.40	12.76	9.32	
	HDD50	10167	2064	1615	1254	588	165	8	1	8	192	736	1493	2043	
	HDD65	14670	2529	2035	1719	1037	594	202	83	211	608	1201	1943	2508	
	CDD50	1029	0	0	0	0	36	268	419	272	34	0	0	0	
	CDD65	57	0	0	0	0	0	12	36	9	0	0	0	0	
	CDH74	1253	0	0	0	1	30	427	586	204	5	0	0	0	
	CDH80	258	0	0	0	0	3	100	127	28	0	0	0	0	
Wind	WSAvg	3.5	2.2	2.5	4.0	5.4	5.6	3.8	3.3	3.2	3.8	3.7	2.4	2.1	
Precipitation	PrecAvg	17.0	0.2	0.2	0.2	0.7	1.3	3.0	4.5	3.9	1.9	0.6	0.4	0.2	
	PrecMax	25.2	0.4	0.7	0.7	1.7	4.3	7.3	9.1	9.3	3.9	2.1	0.9	0.9	
	PrecMin	9.8	0.0	0.0	0.0	0.0	0.0	0.5	1.3	0.7	0.4	0.0	0.0	0.0	
	PrecStd	3.6	0.1	0.2	0.2	0.4	0.8	1.7	2.0	1.8	1.0	0.4	0.2	0.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.3	28.7	46.7	65.5	78.6	87.5	87.5	83.9	74.0	58.3	36.8	16.7	
		MCWB	8.6	23.0	36.2	48.4	58.0	65.7	66.4	65.7	58.2	45.4	30.4	14.4	
	2%	DB	6.7	21.3	40.1	56.7	72.6	82.8	83.8	79.4	68.8	51.1	29.5	8.1	
		MCWB	4.9	17.5	31.7	42.8	53.3	64.2	65.8	63.7	55.4	41.2	25.2	6.7	
	5%	DB	2.3	15.4	35.0	51.5	68.0	78.8	80.4	75.7	64.0	45.6	24.2	3.4	
		MCWB	0.9	12.8	28.2	39.8	51.2	61.6	64.8	63.0	52.7	37.4	21.3	2.3	
	10%	DB	-1.7	10.5	30.2	46.7	63.3	75.2	77.1	72.2	59.7	40.9	19.3	-1.2	
		MCWB	-2.7	8.3	25.1	36.4	48.8	60.0	63.8	61.1	50.6	34.6	17.0	-2.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.9	23.2	36.4	49.0	58.4	71.1	70.9	68.9	60.7	45.6	30.9	14.3
			MCDB	11.2	28.3	46.1	63.5	75.3	79.4	80.7	76.9	69.6	57.1	36.5	16.3
2%		WB	4.9	17.7	32.0	43.7	54.8	66.5	68.7	66.5	57.7	41.4	25.6	6.6	
		MCDB	6.7	21.1	39.7	56.2	69.2	78.2	78.5	75.0	64.9	50.2	29.2	7.9	
5%		WB	1.0	12.9	28.5	40.1	52.4	63.7	66.8	64.6	54.7	38.1	21.5	2.3	
		MCDB	2.3	15.4	34.3	50.5	64.6	75.2	75.8	72.4	62.0	44.7	24.0	3.3	
10%		WB	-2.6	8.4	25.2	37.1	49.7	61.5	65.1	62.6	51.6	34.8	17.1	-2.1	
		MCDB	-1.7	10.3	30.1	46.0	61.7	72.5	73.7	70.4	58.4	40.2	19.3	-1.2	
Mean Daily Temperature Range		5% DB	MDBR	24.1	31.6	30.4	25.0	25.7	27.3	23.5	23.5	24.2	22.7	23.9	21.4
			MCDBR	24.7	34.5	32.6	34.9	37.5	36.2	32.0	31.3	32.8	29.2	27.0	23.6
	MCWBR		23.2	30.9	24.8	22.0	20.2	18.5	16.1	17.0	19.5	19.7	22.5	22.1	
	5% WB	MCDBR	24.8	34.2	31.1	33.0	32.8	30.8	25.4	25.0	26.4	28.0	26.1	23.4	
		MCWBR	23.3	30.6	24.1	21.2	18.4	16.7	14.0	14.5	16.5	19.1	21.8	21.9	
Clear-Sky Solar Irradiance	taub	0.239	0.265	0.304	0.402	0.491	0.459	0.425	0.391	0.339	0.304	0.263	0.243		
	taud	2.194	2.180	2.147	2.048	1.904	2.113	2.272	2.348	2.442	2.412	2.250	2.121		
	Ebn at Noon	235	265	276	259	241	252	260	263	265	250	221	202		
	Edh at Noon	19	28	36	47	57	47	39	35	28	23	18	17		
All-Sky Solar Radiation	RadAvg	387	746	1288	1688	1674	1838	1678	1426	1078	724	435	289		
	RadStd	11	16	39	98	123	121	128	121	82	44	20	12		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page